Optics Letters

Stability of silicon resonator temperature sensors with the Pound–Drever–Hall technique

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Received 25 October 2022; revised 6 January 2023; accepted 6 January 2023; posted 6 January 2023; published 7 February 2023

In this paper, we research the temperature stability of silicon-based ring resonator thermometers utilizing the Pound–Drever–Hall (PDH) technique. A slight temperature fluctuation of 12.2 mK in 200 s was experimentally detected by immersing the sensor in the triple point of water (TPW) system with ultrahigh precision. Additionally, factors that affect temperature stability, including fundamental thermal noise, laser frequency drift, and power fluctuation were analyzed and calculated theoretically. This shows high consistency with experimental results. Moreover, it is proved that the laser drift can be suppressed from 11.3 pm to 0.013 pm with the developed experimental system based on the PDH technique. The silicon-based ring resonator as a potential platform for precise temperature monitoring is proved based on this work. © 2023 Optica Publishing Group

https://doi.org/10.1364/OL.479193

Currently, optical sensors have been widely used in temperature sensing due to their electromagnetic immunity and small size. Different fiber sensor structures based on the fiber Bragg grating (FBG) [1], Mach–Zehnder interferometer (MZI) [2], and Fabry–Perot interferometer (FPI) [3] have been reported. Taking advantage of complementary metal-oxide semiconductor (CMOS) compatibility and high-quality factors, the temperature sensors based on silicon microring resonators (MRRs) have also been demonstrated [4]. Recently, research has been devoted to improving the sensitivity of optical sensors with different methods such as cascaded MRRs to realize the Vernier effect [5] and using the negative thermo-optic coefficient materials as a top cladding [6].

In addition to sensitivity, stability is also a critical characteristic parameter for applying precise temperature measurements. However, the stability of the optical sensors is severely affected by power fluctuations and the output frequency drifts of the incident laser besides the intrinsic noise. Recently, Luo *et al.* [7] demonstrated a stability of 0.8 mK at an averaging time of 2 s by utilizing a self-referenced sensor with a lithium niobate microdisk resonator. By using the Pound–Drever–Hall (PDH) technique, a short-term fluctuation of temperature of 0.16 μK was achieved with a fiber-optic sensor [8]. However, to date, the stability analysis of the silicon-based temperature sensor is still limited to a few seconds. Furthermore, the analysis at thermodynamic temperature reference points is unavailable since the traditional temperature control platforms cannot provide an accurate ambient. Xu *et al.* [9] have proved that at timescales of 1 min, the noise might reach ±5 K, which limits the application in the field of accurate temperature monitoring.

In this paper, we demonstrated a silicon-based MRR temperature sensor with ultrahigh stability utilizing a PDH system. Moreover, its temperature stability was accurately measured by the virtue of the triple point of water (TPW) system. Theoretical analysis and experimental verification of the factors affecting the stability are also carried out separately. To stabilize the laser, the experimental setup based on the PDH technique is established, with which the laser drift was suppressed from 11.3 pm to 0.013 pm. The results show that the temperature fluctuation of the silicon-based sensor is 12.2 mK in 200 s. To the best of our knowledge, the stability of the MRR temperature sensor is first demonstrated and analyzed at the reference point of thermodynamic temperature in detail, and paves a new way for the MRR sensor as a potential platform for precise temperature detection in extreme environments.

The PDH technique as a phase-sensitive method has been used in high-resolution applications [10]. Figure 1 shows the setup of a traditional PDH system and the corresponding frequency spectrum. The phase-modulated laser signal generates two sidebands spaced from the carrier by the modulation frequency, which are detected by a photodiode and yield two heterodyne beats resulting from the beat between the carrier and each sideband. On resonance, the two beats cancel each other out since they were 180° apart in phase. Otherwise, when the laser slightly differs from the resonant frequency, the two beats cannot offset. This will produce the PDH curve after demodulating by a mixer. The curve exhibits a linear region, and its slope indicates the relationship between the error signal and modulating frequency, which is defined as the discriminant coefficient (DC) [11].

Figure 2(a) shows the experimental setup based on the PDH technique for precise temperature measurement. A tunable laser (NEWPORT TLB-6728) with an output from 1520 nm to 1570 nm is used to probe the sensor. The signal modulation was achieved by an electro-optical modulator (THORLABS) and a function generator (TEKTRONIX AFG1062). A mixer (MINI-CIRCUITS ZLW-6) and an amplifier were able to demodulate

Corrected 8 February 2023

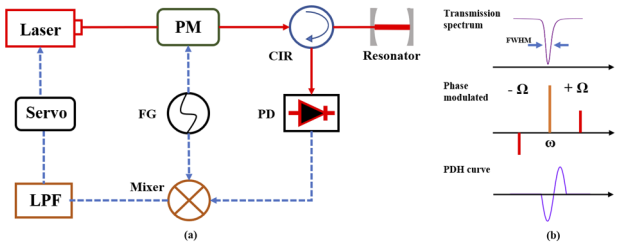

Fig. 1. (a) Configuration and (b) corresponding spectrum of a traditional PDH system. FG: function generator; LPF: low-pass filter.

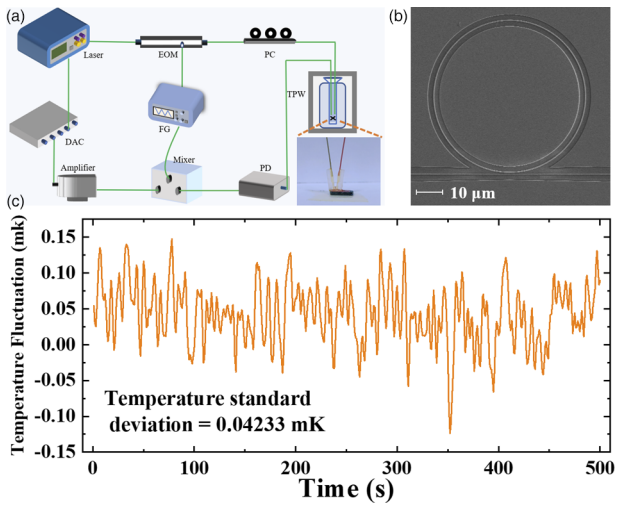

Fig. 2. (a) Schematic diagram of the experimental setup. (b) SEM photo of the MRR sensor. (c) Temperature fluctuation of the TPW.

the signal and obtain the PDH error signal. The data acquisition card (NI USB-6218) was used to collect the demodulated signal and feed the error signals back to the laser to lock the laser output onto the resonant wavelength of the MRR. At the same time, a computer was used to record these error signals reflecting the slight variation of the resonant wavelength. The packaged sensor shown in the inset figure in Fig. 2(a), with a Q value of 53,276 and an FSR of 3.1 nm, was immersed in the TPW (FLUKE-7312) for stability testing. The MRR sensor fabricated on the SOI wafer using standard CMOS technology consists of a 480-nm-wide straight waveguide and a ring with a diameter of 60 μm. Figure 2(b) shows the SEM photo of the device. Based on the traditional method, TPW with a temperature fluctuation standard deviation of 0.04233 mK is demonstrated, and the result was recorded by a commercial standard platinum resistance, as shown in Fig. 2(c).

The high signal-to-noise ratio PDH error signal requires sufficient input power for effective locking [12], thus the laser power was fixed at 2.4 mW. However, the power inside the waveguide is less than 1 mW due to the low coupling efficiency between the fiber and the grating couplers [13]. The sweeping triangular wave signal and the corresponding PDH error signals are plotted in Fig. 3(a). The triangle wave signal, with a frequency of 20 Hz and a peak-to-peak value of 6 V, was applied to the laser to realize a tuning range of ±240 pm centered on the original laser output wavelength as the externally applied voltage of 3 V can bring a 240 pm wavelength tuning to the laser. Therefore,

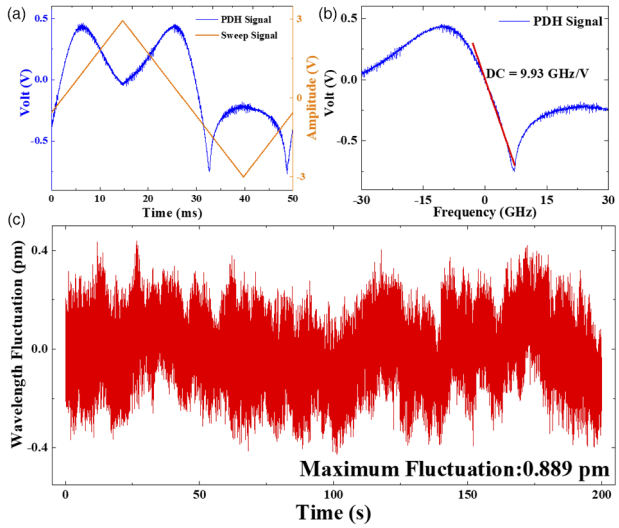

Fig. 3. (a) PDH signal and triangular wave signal. (b) Discriminant coefficient. (c) Resonant wavelength fluctuation of the MRR.

the tuning frequency is 120 GHz in a sweeping cycle. The DC of the PDH curve in this work is defined as the ratio of tuning frequency bandwidth to PDH voltage. The symmetry axis of the error signal in Fig. 3(b) is below the x axis since transmission loss of the electrical signal is unavoidable. However, the frequency is still locked at the symmetry point of the PDH curve, so it has no impact on the experiment results. In Fig. 3(c), the resonant wavelength fluctuation of the MRR sensor was recorded in real time for 200 s, and the result shows the maximum drift is 0.889 pm, corresponding to a temperature fluctuation of 12.2 mK as the sensitivity of the device is 72.89 pm/K. The result shows the high stability of the MRR temperature sensor for a long time.

According to theoretical analysis, the factors influencing the stability mainly arise from four parts [9]: (a) fundamental thermal noise; (b) drift of the laser frequency; (c) power fluctuation of the laser; and (d) temperature fluctuation of the experimental environment. The noise induced by the environment can be beneficially excluded from the TPW system with an ultrahigh temperature accuracy of 0.04233 mK. In this work, we present an in-depth investigation on the former three factors.

For high-precision temperature measurement, fundamental noise derived from the intrinsic thermal variations of silicon for the developed silicon temperature sensor cannot be neglected, which mainly include thermo-refractive noise, thermo-elastic noise, and thermal expansion noise. Thermo-refractive noise, due to the thermal motion of silicon molecules, can be obtained from the well-known thermodynamic equation [14]:

$$<\Delta T^2> = \frac{k_B T^2}{C_P V \rho}, \tag{1}$$

where T is the environment temperature, k_B is the Boltzmann constant, C_p and ρ are the specific heat capacity and the density of the resonator host material, respectively, and V is the volume of the device. Its value is calculated to be 0.032 mK for the developed device. For the thermal expansion noise, one can usually ignore it since the coefficient of thermal expansion (CTE) of silicon (2.7×10^{-6} K) is smaller than the thermo-optic coefficient (CTO) by two orders of magnitude (1.8×10^{-4} K).

However, the thermo-elastic noise induced by the volume fluctuation should be taken into account due to the small volume of the MRR. The wavelength fluctuation resulting from these two noise sources can be expressed as [15]

$$\left(\frac{\delta\lambda}{\lambda}\right)^2 = k_B T \frac{\beta_T}{9V} + \alpha_L^2 \frac{k_B T^2}{C_P V \rho}, \tag{2}$$

where $\delta\lambda$ is the wavelength fluctuation, λ is the center wavelength of the laser, β_T is the isothermal compressibility of the material, and α_L is the linear CTE. The relative calculation results are 0.165 mK and 0.005 mK, which refer to the first and second terms on the right side of Eq. (2), respectively. This demonstrates the fact that the thermal expansion noise can be neglected.

The drift of laser output wavelength induced by the self-heating effect severely constrains precise sensing of the optical applications. To solve this problem, the experimental setup based on the PDH technique was built-up to stabilize the wavelength. Furthermore, a temperature-insensitive HCN gas chamber was used to quantify the wavelength fluctuation of the system. The wavelength fluctuation of the conventional experimental setup and developed system were recorded in Figs. 4(a) and 4(b), respectively, for comparison. Figure 4(a) shows the laser drift of 11.3 pm without laser stabilization, which will critically affect the sensing stability and accuracy. Figure 4(b) shows that the wavelength fluctuation has significantly reduced to 0.013 pm with the stabilization, reflecting a temperature fluctuation of 0.178 mK. It can be seen that the impact of laser drift on stability is suppressed greatly with the constructed experimental setup.

Since the PDH technique is a phase-sensitive method, the impact on the experimental system due to the laser power fluctuation can be ruled out. The main impact of the power fluctuation comes from the resonant wavelength shift induced by the non-linear effects of the ring resonator. The high-Q and small mode volume of the silicon-based MRR result in a strong resonant enhancement of the light, thus it will produce a larger circulating intensity in the ring even for a small input power [16].

The non-linear effects will result in a wavelength shift and the distortion of the spectral response. The main mechanisms are two-photon-absorption (TPA), the Kerr effect (KE), free-carrier dispersion (FCD), and free-carrier absorption (FCA) [17]. When the incident photon energy is larger than half of the indirect energy bandgap of silicon, there exits a TPA effect which generates free carriers [18]. The carrier concentration is related to the power:

$$N = \frac{\beta_{TPA} \tau \lambda P_c^2}{2hcA_{eff}^2}, \tag{3}$$

where β_{TPA} is the TPA absorption coefficient, τ is the effective recombination lifetime of free carriers, and P_c is the cavity power

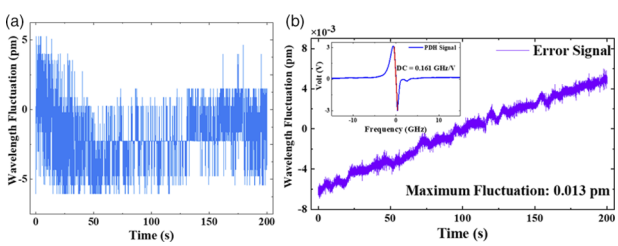

Fig. 4. Laser wavelength drifts (a) without and (b) with stabilization.

within the ring. P_c can be expressed as [17]

$$P_c = \frac{\kappa^2(1-\eta^2)}{|1 - t^2 a e^{j\theta}|^2} P_{in}, \tag{4}$$

where P_{in} is the pump power; η, κ, and t are the coupling loss, coupling coefficient, and transmission coefficient between the bus ring waveguide, respectively; and a is the modal loss of the optical field per round trip. The free carriers resulting from the TPA effect raise the overall optical loss and cause a change in the refractive index (KE and FCD). The former reduces the quality factor, while the latter causes a fluctuation in the resonant wavelength. Non-linear loss due to TPA is expressed as [19]

$$\alpha_{TPA} = \frac{\beta_{TPA}}{A_{eff}} P_c. \tag{5}$$

Fluctuation of the wavelength caused by the KE and FCD are

$$\delta\lambda = \frac{n_2 \lambda P_c}{A_{eff} n} \tag{6}$$

$$\delta\lambda = -\frac{q^2\lambda^3}{2\varepsilon_0 n^2 c^2}\left(\frac{N_e}{m_e^*} + \frac{N_h}{m_h^*}\right), \tag{7}$$

where n_2 is the Kerr nonlinearity coefficient, and m_e^* and m_h^* are the effective mass of the electron and hole, respectively. The modal loss introduced by the FCA is expressed as a function of the free carriers, through an empirical expression [20]:

$$\alpha_{FC} = \frac{q^3\lambda^2}{\varepsilon_0 n c^3}\left(\frac{N_e}{\mu_e m_e^{*2}} + \frac{N_h}{\mu_h m_h^{*2}}\right). \tag{8}$$

Recombination of the free carriers will release energy in the form of heat, which leads to the redshift of the resonant wavelength:

$$\delta\lambda = \Gamma \frac{dn_{Si}}{dT} \frac{Z_T \lambda}{n} P_c (1 - e^{-(\alpha_0 + \alpha_{TPA} + \alpha_{FC})L}). \tag{9}$$

Here, Γ, Z_T, α_0, and L represent the optical confinement factor, the thermal impedance, the linear loss, and the circumference of the ring, respectively. In this equation, the bend loss of the ring waveguide is neglected.

According to the above analysis, we have calculated the relationship between resonant wavelength and cavity power induced by different non-linear effects (the parameters for theoretical calculation are shown in Supplement 1), as shown in Fig. 5(a). The results show that the self-heating effects are dominant, and the FCD is also important. The theoretical and experimental results of the overall impact caused by the non-linear effects on the resonant wavelength are plotted in Fig. 5(b). It can be seen that a small laser output power change will lead to a significant fluctuation of the resonant wavelength. With the developed system, we found that the temperature increased by 0.15 K when laser power increased from 2.35 mW to 2.45 mW, as shown in Fig. 5(c). This means that the ratio of temperature-to-power is about 1.5 K/mW. Figure 5(d) shows the fluctuation of the laser power is about 5.72 μW when applied power is fixed at 2.4 mW. Therefore, the corresponding temperature fluctuation is about 8.58 mK.

Lastly, we analyze the instability of the error signal with the Allan deviation method. Allan deviation is a time-domain measure of frequency stability, which can recognize the different noise sources by examining the slope at different time constants [21]. It can be seen from Fig. 6 that the developed system is

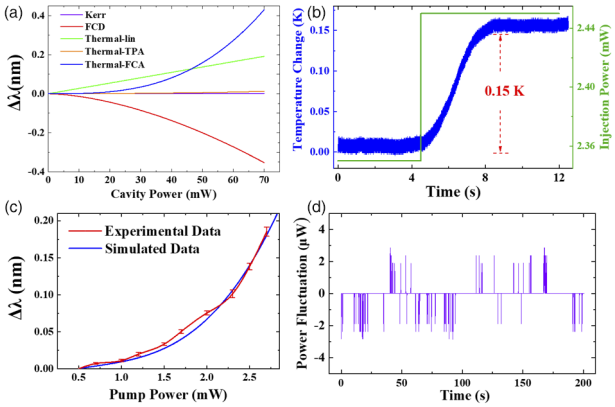

Fig. 5. (a) Different non-linear effect contributions to the wavelength drift as a function of the cavity power. (b) Theory and experimental results of overall non-linear impact on the resonant wavelength. (c) Non-linear effect of the silicon MRR temperature sensor. (d) Power fluctuation of the laser.

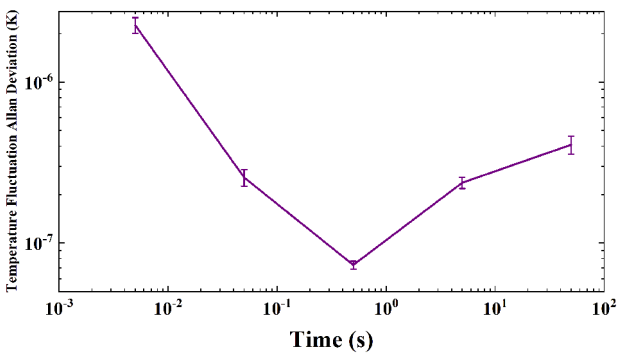

Fig. 6. Allan deviation of the measured error signal fluctuations.

limited by the white noise within 0.5 s and the random walk frequency noise within 5 s, respectively. Later, the slope leveled off, which is mainly influenced by external noise like frequency drift and laser power fluctuation. The stability level eventually reaches about 3.5×10^{-7} K, demonstrating the ability to measure the temperature fluctuation with precision approaching the μK level based on our system.

In conclusion, the stability of the silicon-based MRR temperature sensor is studied with the TPW system. By utilizing a servo-loop locking system based on the PDH technique, we first demonstrated its temperature fluctuation is 12.2 mK in 200 s. By fully studying the factors influencing stability, the theoretical analysis on temperature stability was conducted and experimentally verified. This work proves that the MRR temperature sensor can be used as a potential platform for precise temperature measurements. The dominant element arising from the power drift due to the strong non-linear effect of silicon-based MRRs can be reduced with power stabilization techniques for further study.

Funding. National Key Research and Development Program of China (2022ZD0208601); Strategic Priority Research Program of Chinese Academy of Sciences (XDA25040100, XDA25040200, XDA25040300); National Natural Science Foundation of China (42127807-03); Shanghai Municipal Science and Technology Major Project (2021SHZDZX); Shanghai Pilot Program for Basic Research - Shanghai Jiao Tong University (21TQ1400203); Oceanic Interdisciplinary Program of Shanghai Jiao Tong University (SL2020MS017, SL2103); Scientific Research Fund of Second Institute of Oceanography, MNR (SL2020ZD205).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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